STAT6611
Lecture 03 (T, Sept 1)

- No homework yet (assign R + then due in 2 weeks)
- Office hours soon

Lemma 1.1. *A matrix G is a generalized inverse of a matrix A if and only if G' is a generalized inverse of A', where A' is the transpose of A.*

pf: $AGA = A \Leftrightarrow A' = (AGA)' = A'G'A'$

G g.i of A — G' g.i of A'

Corollary. If A is a symmetric matrix, then $(A^-)'$ is a generalized inverse of A. In particular, for any matrix X, $[(X'X)^-]'$ is a generalized inverse of $X'X$.

pf: $(A^-)'$ is g.i of $A' = A$ ↑ If A is symmetric

If $(X'X)^-$ is g.i of $X'X$ then $((X'X)^-)'$ is g.i. too ← (*)

Remark. A generalized inverse of a symmetric matrix A can be nonsymmetric.

e.g. $A = O_{m\times m}$ is symmetric but A^- could be any $m\times m$ matrix

If A is symmetric, we could start from a g.i. A^- & make a new & symmetric g.i $(A^- + (A^-)')/2$

Lemma 1.2. *For any matrix X, $\mathrm{tr}(X'X) = 0$ implies $X = 0$.* ↑ "trace"

Proof. Let $\underline{x}_j$ be the jth column of X. Note that

$$X'X = (\underline{x}_1, \ldots, \underline{x}_n)'(\underline{x}_1, \ldots, \underline{x}_n) = \begin{pmatrix} \underline{x}_1' \\ \vdots \\ \underline{x}_n' \end{pmatrix} (\underline{x}_1, \ldots, \underline{x}_n) = \begin{bmatrix} \underline{x}_1'\underline{x}_1 & \cdots & \underline{x}_1'\underline{x}_n \\ \vdots & & \vdots \\ \underline{x}_n'\underline{x}_1 & \cdots & \underline{x}_n'\underline{x}_n \end{bmatrix}.$$

(X' and X)

& $\mathrm{tr}(X'X) = \sum_{i=1}^n \underline{x}_i'\underline{x}_i = \sum_{i=1}^n \|\underline{x}_i\|^2$

So $\mathrm{tr}(X'X) = 0 \Rightarrow$ each $\|\underline{x}_i\|^2 = 0$ $i = 1..n$

$\Rightarrow$ each $\underline{x}_i = 0$ $i = 1..n \Rightarrow X = 0$

Lemma 1.3. *For any matrix X and for any matrices P and Q of appropriate dimensions, we have*

→ *(i) $PX'X = QX'X$ implies $PX' = QX'$ and*

(ii) $X'XP = X'XQ$ implies $XP = XQ$.

("scratch-off" result)

$XXA = XXB \Rightarrow XA = XB$

$AX'X = BX'X \Rightarrow AX' = BX'$

suppose

Proof of (i): $PX'X = QX'X$ implies that

$$0 = (PX'X - QX'X)(P-Q)' = (XP'-XQ')'X(P'-Q') = (XP'-XQ')'(XP'-XQ').$$

difference $= \tilde{X}'$ $\tilde{X}$ say

$\Rightarrow tr(\tilde{X}'\tilde{X}) = 0$

$\Rightarrow \tilde{X} = 0$ by last result

$\Rightarrow XP' = XQ' \Leftrightarrow PX' = QX'$

g.i of X'

Lemma 1.4. *For any matrix X, $X'X(X'X)^-X' = X'$, that is, $X(X'X)^-$ is a generalized inverse of X'.*

proof: $(X'X)(X'X)^-(X'X) = X'X$ by def

$\Rightarrow (X'X)(X'X)^-X' = X'$ Lemma 1.3

$\Rightarrow X(X'X)^-$ is g.i of X' by def

$(X'X)^-X'$ is g.i. of X

Corollary 1. (i) $X'X[(X'X)^-]'X' = X'$, (ii) $X(X'X)^-X'X = X$, and (iii) $X[(X'X)^-]'X'X = X$.

$X[(X'X)^-]'$ is g.i. of X'

$((X'X)^-)'X'$ is g.i of X

Note Lemma 1.4 says "$X \cdot$ g.i of $(X'X)^-$" is a g.i. of X'

But for any g.i of $(X'X)$ (say $(X'X)^-$) it holds that

$((X'X)^-)'$ is also a g.i. of $(X'X)$ $\Rightarrow X((X'X)^-)'$ is a g.i of X' (shows (i))

$\Rightarrow (X'X)^-X'$ is a g.i. of X by Lemma 1.1 (shows (ii))

(iii) follows from (ii) since $((X'X)^-)'$ is g.i of $(X'X)$ so $((X'X)^-)'X'$ is g.i of X

Corollary 2. Corresponding to any matrix X, there exists a matrix B satisfying $X'XB = X'$.

important

i.e. take $B = (X'X)^-X'$ or $((X'X)^-)'X'$ by Lemma 1.4/ Corollary 1

Lemma 1.5. *Let X be any matrix and let B_1 and B_2 be any two matrices that satisfy $X'XB = X'$. Then, $XB_1 = XB_2$.*

proof: if B_1 & B_2 satisfy $X'XB_1 = X' = X'XB_2$
then $XB_1 = XB_2$ holds by Lemma 1.3 (scratch-off)

Insight: if B_1 solves $X'XB_1 = X'$, then B_1 acts like $B_2 = (X'X)^-X'$ or $((X'X)^-)'X'$ (both solve $X'XB_2 = X'$) when computing $XB_1 = XB_2 = X(X'X)^-X' = X((X'X)^-)'X'$

Corollary 1. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, $XBX = B'X'X = X$.

proof: $XB = X(X'X)^-X'$ by Lemma 1.5 ($B_1 = B$ & $B_2 = (X'X)^-X'$)

$\Rightarrow XBX = X(X'X)^-X'X = X$ by Cor 1/Lemma 1.4

($(X'X)^-$ g.i. of $X'X$)

$\Rightarrow B'X'X = (X(X'X)^-X')'X = X((X'X)^-)'X'X = X$

(B' g.i. of X; $((X'X)^-)'$ g.i. of X by Cor 1/Lem 1.4)

Corollary 2. Let X be any matrix and let G_1 and G_2 be any two generalized inverses of $X'X$. Then, $XG_1X' = XG_2X'$.

proof: Note $B_1 = G_1X'$ or $B_2 = G_2X'$ solves $X'XB = X'$ by Lem 1.4

So, $XB_1 = XB_2$ by Lem 1.5

$XG_1X' = XG_2X'$ (G_1, G_2)

Corollary 3. For any matrix X, $X(X'X)^-X' = X[(X'X)^-]'X'$, that is, $X(X'X)^-X'$ is symmetric.

Both $G_1 = (X'X)^-$ & $G_2 = ((X'X)^-)'$ are g.i. of $(X'X)$

Follows by Cor 2 above

Corollary 4. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, the matrix XB is symmetric.

pf: Both $B_1 = (X'X)^- X'$ and $B_2 = ((X'X)^-)' X'$ are solutions too

So $XB = XB_1 = XB_2 = (XB_1)' = X(X'X)^- X'$ is symmetric

Corollary 5. Let $P_X = XB$ where B is a solution to $X'XB = X'$. Then, P_X has the following properties:

(1) P_X is invariant to the choice of solution B. Lem 1.5

(2) P_X is symmetric. Cor 4 / Lem 1.5

(3) $P_X = X(X'X)^- X'$. by (1) & $B = (X'X)^- X'$ is a solution

(4) $P_X X = X$. by Cor 1 / Lem 1.5 ($\underbrace{XBX}_{P_X} = X$)

(5) P_X is idempotent, that is, $P_X^2 = P_X$

↑ by (3)+(4), we have

$P_X^2 = P_X P_X = P_X(X(X'X)^- X')$

$P_X X = X \rightarrow = X(X'X)^- X' = P_X$

Definition. If $P_X = XB$ where B is a solution to $X'XB = X'$, then P_X is said to be an orthogonal projection matrix.

Definition. The column space of an $m \times n$ matrix A consists of all $m \times 1$ vectors spanned by the columns of A, that is, a vector $\underline{c}$ belongs to the column space of A if and only if there exists a vector $\underline{b}$ for which $\underline{c} = A\underline{b}$.

set of all vectors in $\mathbb{R}^m$ as linear combinations of the n columns of A (each in $\mathbb{R}^m$)

Similarly, the row space of A consists of all $1 \times n$ vectors spanned by the rows of A.

Notation. We use $\mathcal{C}(A)$ to denote the column space of A and use $\mathcal{R}(A)$ to denote the row space of A.

Lemma 1.6. *(i) $\mathcal{C}(B)$ is a subspace of $\mathcal{C}(A)$ if and only if $B = AC$ for some matrix C. (ii) $\mathcal{R}(B)$ is a subspace of $\mathcal{R}(A)$ if and only if $B = CA$ for some matrix C.*

Various special cases of the following two theorems are very useful in linear-model theory and methodology.

Theorem 1.4. *Let* $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ *be a partitioned matrix.*

(i) If $\mathcal{C}(C)$ *is a subspace of* $\mathcal{C}(A)$ *and* $\mathcal{R}(D)$ *is a subspace of* $\mathcal{R}(A)$*, then the partitioned matrix*

$$\begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix}$$

is a generalized inverse of M*, where* $Q = B - DA^-C$*.*

(ii) If $\mathcal{C}(D)$ *is a subspace of* $\mathcal{C}(B)$ *and* $\mathcal{R}(C)$ *is a subspace of* $\mathcal{R}(B)$*, then the partitioned matrix*

$$\begin{bmatrix} P^- & -P^-CB^- \\ -B^-DP^- & B^- + B^-DP^-CB^- \end{bmatrix}$$

is a generalized inverse of M*, where* $P = A - CB^-D$*.*

Theorem 1.5. *Let* $W = R + D$*, where* $\mathcal{C}(D)$ *is a subspace of* $\mathcal{C}(R)$ *and* $\mathcal{R}(D)$ *is a subspace of* $\mathcal{R}(R)$ *(as would be the case, for example, if* R *were nonsingular). Take* S, T*, and* U *to be any three matrices such that*

$$D = STU.$$

Then, the matrix

$$R^- - R^-ST(T + TUR^-ST)^-TUR^-$$

is a generalized inverse of W*.*

2 Classical (Fixed-Effects) Estimation Theory

2.1 Linear Models

Let $\underline{y}$ represent an $n \times 1$ vector of (scalar-valued) data points. We suppose that

$$\underline{y} = X\underline{\beta} + \underline{\epsilon},$$

where X is a given $n \times p$ "design" matrix, $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters that is contained in some given parameter space, and $\underline{\epsilon}$ is an $n \times 1$ vector of unobservable "residuals" or "deviations."

It is convenient to introduce various special cases of this "linear model."

Model I (Least squares model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; parameter space $= \{\underline{\beta} : \underline{\beta} \in \mathbb{R}^p\}$ (where $\mathbb{R}^p$ denotes the Euclidean p-space). ↑ no assumptions on $\underline{\epsilon}$ (most general model)

Model II (Gauss-Markov model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 I$; parameter space $= \{\underline{\beta}, \sigma^2 : \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\}$.

Model III (Aitken model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given positive definite or positive semidefinite matrix; parameter space $= \{\underline{\beta}, \sigma^2 : \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\}$. if $H = I$, get model II

Model IV (Mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$;

$$\underline{\epsilon} = Z_1\underline{b}_1 + \cdots + Z_m\underline{b}_m + \underline{e},$$

where Z_i is a given $n \times q_i$ "design" matrix, $\underline{b}_i$ is a $q_i \times 1$ random vector such that $E(\underline{b}_i) = \underline{0}$ and $\mathrm{Var}(\underline{b}_i) = \sigma_i^2 I$, and $\underline{e}$ is an $n \times 1$ random vector such that $E(\underline{e}) = \underline{0}$ and $\mathrm{Var}(\underline{e}) = \sigma^2 I$, and where $E(\underline{b}_i \underline{e}') = 0$ and $E(\underline{b}_i \underline{b}_j') = 0$ for $i = 1, \ldots, m$, $j = 1, \ldots, m$, and $i \neq j$; parameter space $= \{\underline{\beta}, \sigma_1^2, \ldots, \sigma_m^2, \sigma^2 : \underline{\beta} \in \mathbb{R}^p,\ \sigma_1^2 \geq 0, \ldots, \sigma_m^2 \geq 0,\ \sigma^2 > 0\}$. Here, $\sigma_1^2, \ldots, \sigma_m^2$, and σ^2 are called variance components.

if $\sigma_1^2 = \cdots = \sigma_m^2 = 0$, get model II

Model V (Random-effects model): a special case of the mixed-effects model where $X = \underline{1}$ (a column vector of 1's).

Model VI (General mixed-effects model, with a linear variance structure): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \theta_1 H_1 + \cdots + \theta_m H_m$, where $H_1, \ldots, H_m$ are given symmetric matrices; parameter space $= \left\{\underline{\beta}, \underline{\theta} \colon \underline{\beta} \in \mathbb{R}^p, \underline{\theta} \in \Omega\right\}$, where $\underline{\theta} = (\theta_1, \ldots, \theta_m)'$ and Ω is the set of $\underline{\theta}$-vectors for which $\mathrm{Var}(\underline{\epsilon})$ is positive definite or positive semidefinite, or Ω is a given subset of that set.

Model VII (General mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = V$, where the elements of V are functions of some m-dimensional parameter vector $\underline{\theta}$; parameter space $= \left\{\underline{\beta}, \underline{\theta} \colon \underline{\beta} \in \mathbb{R}^p, \underline{\theta} \in \Omega\right\}$, where Ω is the set of $\underline{\theta}$-vectors for which V is positive definite or positive semidefinite, or Ω is a given subset of that set.

Remarks

- Model II is a special case of Model III.
- Models III and IV (and V) are special cases of Model VI.
- Model VI is a special case of Model VII.
- Models II—VII can all be viewed as special cases of Model I.

In all of Models I—VII, the space of possible values for $\underline{\beta}$ is the Euclidean p-space. Limiting $\underline{\beta}$ to some subset of $\mathbb{R}^p$ gives a constrained version of each model. The kinds of limitations on $\underline{\beta}$ that are most commonly imposed are those determined by linear equality constraints or linear inequality constraints, such as $A\underline{\beta} = \underline{b}$ or $A\underline{\beta} \geq \underline{b}$.

design matrix X is $n \times p$

Notation. Let $p^* = \text{rank}(X)$.

We have $p^* \leq p$. We do not require that $p^* = p$. When $p^* = p$, the linear model is customarily referred to as a full-rank linear model.

The problem to be considered is that of making inferences about the parameters of the linear model or about various functions of these parameters. In particular, we will consider making inferences about linear combinations of the elements of $\underline{\beta}$, such as obtaining a point estimate or confidence interval for a linear combination $\underline{\lambda}'\underline{\beta}$, or testing the null hypothesis $\underline{\lambda}'\underline{\beta} = c$, where c is a specified constant. More generally, we may seek an estimator or a confidence region for a vector $\Lambda'\underline{\beta}$, where Λ is a given $p \times r$ matrix (so that $\Lambda'\underline{\beta}$ is an r-dimensional vector of linear functions of $\underline{\beta}$), or a simultaneous test of the null hypothesis $\Lambda'\underline{\beta} = \underline{c}$, where $\underline{c}$ is a specified vector.

A related problem is that of making inferences about $\underline{\lambda}'\underline{\beta} + u$, where u is a random variable. For example, $\underline{\lambda}'\underline{\beta} + u$ may represent the model for a future data point. The problem is simplest when u is assumed to be uncorrelated with the elements of $\underline{\epsilon}$. We may wish to predict (or estimate) $\underline{\lambda}'\underline{\beta} + u$ or to establish a prediction (or confidence) interval for it. A generalization of this problem is that of simultaneous inference for a vector $\Lambda'\underline{\beta}+\underline{u}$, where $\underline{u}$ is a random vector. We may seek a prediction (or confidence) region for $\Lambda'\underline{\beta} + \underline{u}$.

2.2 Estimability

Definition. An estimator $t(\underline{y})$ of a linear function $\underline{\lambda}'\underline{\beta}$ is said to be linear if $t(\underline{y}) \equiv c + \underline{a}'\underline{y}$ for some constant c and some vector $\underline{a}$ of constants.

Context: you get pick $\underline{\lambda}'$ & $\underline{\beta}$ is unknown parameter (in $\mathbb{R}^p$) & $\underline{\lambda}'\underline{\beta}$ is parametric function"

Definition. Under Models II—VII, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] \equiv \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$ ← regardless of $\underline{\beta} \in \mathbb{R}^p$

Note that, under Models II—VII, $E(c + \underline{a}'\underline{y}) =$ $c + \underline{a}'E(\underline{Y}) = c + \underline{a}'X\underline{\beta}$ for any $c \in \mathbb{R}$ & $\underline{a} \in \mathbb{R}^n$

$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, $E(\underline{\varepsilon}) = 0$

Lemma 2.1. *Under Models II—VII, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.*

⇐ assume $\underline{\lambda}' = \underline{a}'X$ & $c = 0$

$E(c + \underline{a}'\underline{Y}) = \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ for any $\underline{\beta}$ ⇒ $\underline{a}'\underline{Y}$ is unbiased

⇒ If $c + \underline{a}'\underline{Y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$

then $\underline{\lambda}'\underline{\beta} = E(c + \underline{a}'\underline{Y}) = c + \underline{a}'X\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$

If $\underline{\beta} = 0$, then $c = 0$

If pick $\underline{\beta}$ to be columns of $I_{p\times p}$, get $\underline{a}'X I_{p\times p} = \underline{\lambda}' I_{p\times p}$

$\underline{a}'X = \underline{\lambda}'$

Definition. Under Models II—VII, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Otherwise, it is said to be nonestimable.